

Unifying Special Relativity and Quantum Mechanics

Lecture 18 — The Graviton Propagator and Forces from Particle Exchange

(Harmonic gauge, the propagator in D dimensions,
and the vacuum-energy method for potentials)

Typed version of the handwritten notes, with added explanations in blue boxes

How to read this document

Conventions. Exactly as in Lectures 1–17: normal text and **yellow** / **pink** / **grey** boxes are a faithful transcription of the handwritten notes; text colours follow the pens of the original. Everything in a **blue box like this one** is extra material I added.

Pagination. These sheets continue the gravity chapter opened in Lecture 16, so they still carry **nine stars**: [p. 27*****]–[p. 47*****].

Three anomalies in this scan.

- (a) **Overlap at the start.** The first two sheets, [p. 27*****] and [p. 28*****], are a *repeat*: they are the last two sheets of the Lecture 17 scan. They are reproduced here in condensed form (Section 1) so that the sentence Lecture 17 broke off in — “to extract the propagator, we need. . .” — is completed on the page where it belongs.
- (b) **Two sheets out of order.** In the scan, [p. 42*****] comes *before* [p. 41*****]. The logical order is $40 \rightarrow 41 \rightarrow 42 \rightarrow 43$, and that is the order used here: p. 41***** introduces the interaction-picture evolution operator, p. 42***** inserts the resolution of the identity. If you follow the paper scan, expect to flip back one sheet at that point.
- (c) **The scan stops in the middle of the punchline.** Sheet [p. 47*****] sets up the spin-2 exchange between two static masses — and stops before computing it. Section 14 transcribes the setup, and a blue box carries the calculation through to $E_0 = -GM_1M_2/r$, since that result is the entire point of the preceding twenty pages.

Where we left off. Lecture 17 derived the soft-graviton theorem and, from it, the universality of the gravitational coupling; its last page observed that the gauge invariance $h^{\mu\nu} \rightarrow h^{\mu\nu} + \partial^\mu \xi^\nu + \partial^\nu \xi^\mu$ makes the kinetic operator non-invertible, so no propagator exists until the gauge is fixed. This lecture fixes it (harmonic gauge), computes the graviton propagator in D spacetime dimensions, and then answers the question the whole construction was aimed at: *what force does exchanging this particle produce?* The method — reading the potential off the shift in the vacuum energy caused by two static sources — is developed first for the photon, where it reproduces Coulomb, and then applied to gravity.

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Part I — Fixing the gauge

(pp. 27*****–32*****)

1 Recap of the overlapping sheets [p. 27*****–28*****]

These two sheets repeat the end of Lecture 17

Sheets [p. 27*****] and [p. 28*****] are identical to the last two sheets of the Lecture 17 scan and are transcribed in full there (Sections 12–14 of that document). In brief:

The soft theorem. In the soft limit,

$$i\mathcal{M}_1 \approx i\mathcal{M}_0 \varepsilon_{\mu\nu}^*(\vec{k}) \left[\sum_{j=1}^m \kappa_j^{\text{FIN}} \frac{k_j^\mu k_j^\nu}{k_j \cdot k} - \sum_{i=1}^n \kappa_i^{\text{IN}} \frac{p_i^\mu p_i^\nu}{p_i \cdot k} \right],$$

and invariance under $\varepsilon_{\mu\nu} \rightarrow \varepsilon_{\mu\nu} + \xi_\mu k_\nu + \xi_\nu k_\mu$ forces $\sum_j \kappa_j^{\text{FIN}} k_j^\mu = \sum_i \kappa_i^{\text{IN}} p_i^\mu$ (★). Combined with momentum conservation $\sum_i p_i^\mu = \sum_j k_j^\mu$ (★★) this can only hold if

$$\kappa_i^{\text{IN}} = \kappa_j^{\text{FIN}} \equiv \kappa, \quad \forall i, j \quad \text{— gravity is universal (the equivalence principle).}$$

And the opening of the present topic. Starting from

$$\mathcal{L}_0 = \frac{1}{2}(\partial_\mu h_{\rho\sigma})(\partial^\mu h^{\rho\sigma}) - (\partial_\mu h_{\nu\rho})(\partial^\nu h^{\mu\rho}) + (\partial_\mu h^{\mu\nu})(\partial_\nu h) - \frac{1}{2}(\partial_\mu h)(\partial^\mu h),$$

which has the gauge invariance $h^{\mu\nu} \rightarrow h^{\mu\nu} + \partial^\mu \xi^\nu + \partial^\nu \xi^\mu$, we cannot derive the propagator without fixing the gauge. The situation is conceptually analogous to QED: $-\frac{1}{4}F_{\mu\nu}F^{\mu\nu}$ is gauge invariant and, to extract the propagator, we need. . . — and *that* is the sentence the next sheet finally completes.

2 The harmonic gauge [p. 29*****]

... first to fix the gauge. A popular choice is the Lorentz gauge, $\partial_\mu A^\mu = 0$.

We need something analogue to the Lorentz gauge also in the case of gravity. Such gauge is called the **harmonic gauge**, which is defined by the condition

$$\partial_\mu \left(h^{\mu\nu} - \frac{1}{2} g^{\mu\nu} h \right) = 0$$

↔ If we introduce the **trace-reverse tensor**

$$\bar{h}_{\mu\nu} = h_{\mu\nu} - \frac{1}{2} g_{\mu\nu} h$$

the harmonic gauge takes the form

$$\partial_\mu \bar{h}^{\mu\nu} = 0$$

Obs: We should show that it is always possible to make a gauge transformation such that, starting from a generic $h^{\mu\nu}$, we go to a new $h'^{\mu\nu} = h^{\mu\nu} + \partial^\mu \xi^\nu + \partial^\nu \xi^\mu$ such that it satisfies the harmonic gauge condition.

↔ Let's discuss this point more quantitatively.

Added explanation (not in the original notes)

Names. This gauge goes by several names in the literature — *harmonic*, *de Donder*, *Lorenz* (by analogy), *Einstein*. All the same condition. “Harmonic” comes from general relativity, where the condition is equivalent to $\square_g x^\mu = 0$: the four coordinate functions are harmonic with respect to the curved d’Alembertian.

Why “trace-reverse”. Take the trace of the definition: $\bar{h} = g^{\mu\nu} \bar{h}_{\mu\nu} = h - \frac{1}{2} \cdot 4 \cdot h = -h$ in four dimensions. So the operation flips the sign of the trace — and applying it twice gives back $h_{\mu\nu}$: $\bar{\bar{h}}_{\mu\nu} = h_{\mu\nu}$. It is an involution, which is why one can freely trade h for $\bar{h}$ at any point in the calculation. (In D dimensions the trace becomes $\bar{h} = (1 - D/2)h$, still an involution.)

The exact parallel with electromagnetism. In QED the Lorenz condition $\partial_\mu A^\mu = 0$ turns the equation of motion $\partial_\nu F^{\nu\mu} = 0$ into the simple wave equation $\square A^\mu = 0$. Here $\partial_\mu \bar{h}^{\mu\nu} = 0$ will turn the six-term equation of motion of p. 32***** into $\square \bar{h}^{\mu\nu} = 0$. In both cases the point of gauge fixing is to kill the cross-derivative terms and leave a pure d’Alembertian — which is exactly what one needs in order to *invert* the operator and get a propagator.

3 The harmonic gauge is always reachable [p. 30*****]

Consider $h'^{\mu\nu} = h^{\mu\nu} + \partial^\mu \xi^\nu + \partial^\nu \xi^\mu$, and impose the condition that $h'^{\mu\nu}$ satisfies the harmonic gauge condition

$$\partial_\mu \left(h'^{\mu\nu} - \frac{1}{2} g^{\mu\nu} h' \right) \stackrel{!}{=} 0$$

This means that

$$\begin{aligned} \partial_\mu \left[h^{\mu\nu} + \partial^\mu \xi^\nu + \partial^\nu \xi^\mu - \frac{1}{2} g^{\mu\nu} (h + 2\partial_\alpha \xi^\alpha) \right] &= \\ \partial_\mu h^{\mu\nu} + \square \xi^\nu + \cancel{\partial^\nu (\partial_\mu \xi^\mu)} - \frac{1}{2} (\partial^\nu h) - \cancel{\partial^\nu (\partial_\alpha \xi^\alpha)} &= 0 \end{aligned}$$

↔ We see that it is enough to choose ξ^μ such that

$$\square \xi^\nu = - \left(\partial_\mu h^{\mu\nu} - \frac{1}{2} \partial^\nu h \right)$$

for the transformed field $h'^{\mu\nu}$ to verify the harmonic gauge condition.

Added explanation (not in the original notes)

Why this settles it. $\square \xi^\nu =$ (given source) is just four decoupled wave equations, one for each ν , and a wave equation with a source always has solutions (retarded, advanced, or any combination). So no matter what $h^{\mu\nu}$ you start from, a gauge transformation into harmonic gauge exists. The gauge is *attainable*, which is the only thing that had to be checked.

Note the trace-reversed form. The right-hand side is $-\partial_\mu \bar{h}^{\mu\nu}$, i.e. $\square \xi^\nu = -\partial_\mu \bar{h}^{\mu\nu}$: the gauge parameter has to absorb exactly the failure of the original field to be transverse. Compare QED, where reaching Lorenz gauge requires $\square \alpha = -\partial_\mu A^\mu$.

4 Residual gauge freedom: from 10 to 6 to 2 [p. 30*****-32*****]

Let us now comment about the propagating DOF. The **harmonic gauge condition** is a set of 4 equations that can be used to **eliminate 4 out of the 10 DOF** contained inside $h^{\mu\nu}$, so that we are left with 6 independent components.

↔ This is not the end of the story. The harmonic gauge admits a **residual gauge freedom**

[p. 31*****]

To see this, let's do a second gauge transformation

$$h'^{\mu\nu} \longrightarrow h''^{\mu\nu} = h'^{\mu\nu} + \partial^\mu \xi'^\nu + \partial^\nu \xi'^\mu$$

such that $h''^{\mu\nu}$ still satisfies the harmonic gauge condition.

This is indeed possible. Let's impose the condition $\partial_\mu (h''^{\mu\nu} - \frac{1}{2} g^{\mu\nu} h'') \stackrel{!}{=} 0$. We get

$$\begin{aligned} & \partial_\mu \left(h'^{\mu\nu} + \partial^\mu \xi'^\nu + \partial^\nu \xi'^\mu - \frac{1}{2} g^{\mu\nu} (h' + 2(\partial_\alpha \xi'^\alpha)) \right) = \\ &= \underbrace{\partial_\mu \left(h'^{\mu\nu} - \frac{1}{2} g^{\mu\nu} h' \right)}_{=0, \text{ since } h'^{\mu\nu} \text{ satisfies the harmonic gauge}} + \left(\square \xi'^\nu + \cancel{\partial^\nu (\partial_\alpha \xi'^\alpha)} - \cancel{\partial^\nu (\partial_\alpha \xi'^\alpha)} \right) = \square \xi'^\nu \stackrel{!}{=} 0 \end{aligned}$$

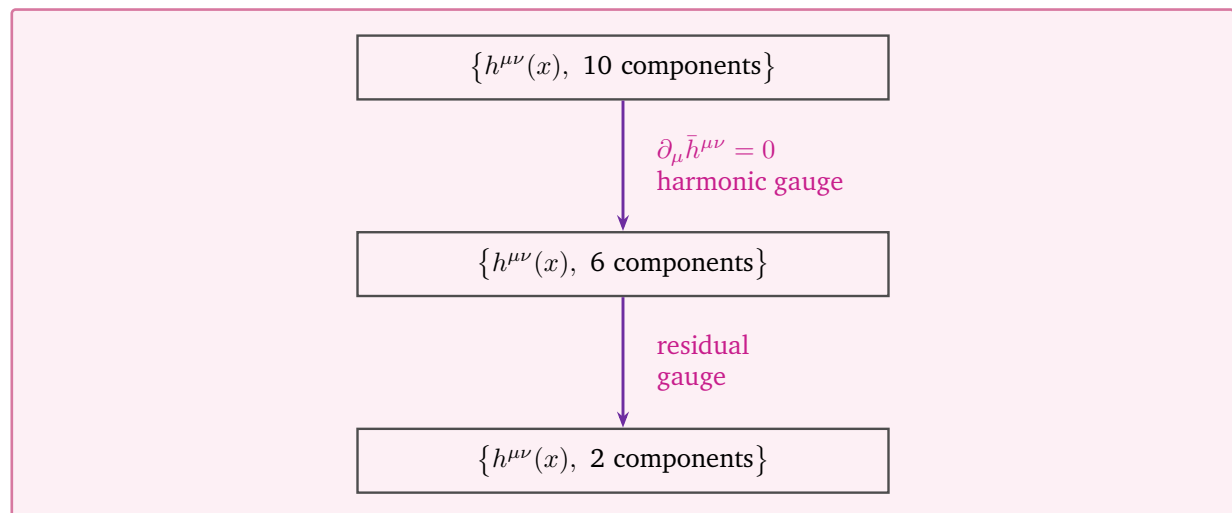
The bottom line is that if we do a second gauge transformation

$$h'^{\mu\nu} \longrightarrow h''^{\mu\nu} = h'^{\mu\nu} + \partial^\mu \xi'^\nu + \partial^\nu \xi'^\mu \quad \text{with} \quad \square \xi'^\mu = 0$$

the field $h''^{\mu\nu}$ still satisfies the harmonic gauge.

$\hookrightarrow$ We can choose the four harmonic functions ξ'^μ (such that $\square \xi'^\mu = 0$) in order to eliminate 4 more components of $h^{\mu\nu}$.

In summary [p. 32*****]



Added explanation (not in the original notes)

Why the residual freedom exists at all. Fixing $\partial_\mu \bar{h}^{\mu\nu} = 0$ does not exhaust the gauge group: it only requires $\square \xi^\nu$ to take a specific value. Any further transformation with $\square \xi'^\nu = 0$ — four *harmonic* functions — preserves the condition. Since a harmonic function is fixed by its initial data, these are genuine functional degrees of freedom, and they can be used to remove four more components.

$10 - 4 - 4 = 2$. That is the graviton's two helicities, ± 2 . Note that these are the *same* two counted in the TT gauge on p. 13***** of Lecture 17 — indeed the residual freedom is precisely what allows one to go from harmonic gauge to the traceless–transverse gauge.

The photon has the identical structure. Lorenz gauge $\partial_\mu A^\mu = 0$ removes one of the four components; the residual transformations with $\square \alpha = 0$ remove one more; $4 - 1 - 1 = 2$, the two photon helicities. Same pattern, one index fewer.

Part II — The graviton propagator

(pp. 32*****-37*****)

5 The equation of motion in the harmonic gauge [p. 32*****-33*****]

We now write the equation of motion in the harmonic gauge.

The E.o.M. are

$$\square h^{\rho\sigma} - \partial^\rho (\partial_\mu h^{\mu\sigma}) - \partial^\sigma (\partial_\mu h^{\mu\rho}) + \partial^\rho \partial^\sigma h + g^{\rho\sigma} \partial_\nu (\partial_\mu h^{\mu\nu}) - g^{\rho\sigma} \square h = 0$$

↪ We use the harmonic gauge condition $\partial_\mu h^{\mu\nu} = \frac{1}{2}(\partial^\nu h)$

$$\square h^{\rho\sigma} - \cancel{\partial^\rho (\frac{1}{2}\partial^\sigma h)} - \cancel{\partial^\sigma (\frac{1}{2}\partial^\rho h)} + \cancel{\partial^\rho \partial^\sigma h} + \underbrace{g^{\rho\sigma} \partial_\nu (\frac{1}{2}\partial^\nu h)}_{\frac{1}{2}g^{\rho\sigma}\square h - g^{\rho\sigma}\square h} - g^{\rho\sigma}\square h = 0$$

We find [p. 33*****]

$$\square h^{\rho\sigma} - \frac{1}{2}g^{\rho\sigma}\square h = 0$$

E.o.M. in the harmonic gauge

Added explanation (not in the original notes)

Check of the three cancellations. With $\partial_\mu h^{\mu\nu} = \frac{1}{2}\partial^\nu h$ the second and third terms become $-\frac{1}{2}\partial^\rho \partial^\sigma h$ each, i.e. $-\partial^\rho \partial^\sigma h$ in total, which cancels the fourth term $+\partial^\rho \partial^\sigma h$ exactly. The fifth becomes $+\frac{1}{2}g^{\rho\sigma}\square h$, which combines with the sixth $-g^{\rho\sigma}\square h$ to give $-\frac{1}{2}g^{\rho\sigma}\square h$.

The result is just the wave equation for $\bar{h}$. Indeed $\square h^{\rho\sigma} - \frac{1}{2}g^{\rho\sigma}\square h = \square \bar{h}^{\rho\sigma}$, so the equation of motion in harmonic gauge reads simply

$$\square \bar{h}^{\rho\sigma} = 0$$

— the graviton is a massless wave, exactly as $\square A^\mu = 0$ for the photon in Lorenz gauge. This is the flat-space, source-free version of the linearized Einstein equations, and it is what gravitational waves solve.

6 Setting up the Green's function [p. 33*****-35*****]

Now that the gauge is fixed, we compute the propagator.

■ First of all, we rewrite the E.o.M. as follows:

$$\begin{aligned} \square h^{\rho\sigma} - \frac{1}{2}g^{\rho\sigma}\square h &= \square (h_{\mu\nu}g^{\mu\rho}g^{\nu\sigma}) - \frac{1}{2}g^{\rho\sigma}\square (g^{\mu\nu}h_{\mu\nu}) = \\ &= g^{\mu\rho}g^{\nu\sigma}\square h_{\mu\nu} - \frac{1}{2}g^{\rho\sigma}g^{\mu\nu}\square h_{\mu\nu} = \\ &= \frac{1}{2}(g^{\mu\rho}g^{\nu\sigma} + g^{\nu\rho}g^{\mu\sigma} - g^{\rho\sigma}g^{\mu\nu})\square h_{\mu\nu} = 0 \end{aligned}$$

$$\frac{1}{2}(g^{\mu\rho}g^{\nu\sigma} + g^{\nu\rho}g^{\mu\sigma} - g^{\rho\sigma}g^{\mu\nu})\square h_{\mu\nu}(x) = 0$$

■ We look for a Green's function of the above operator. We setup the equation

$$\frac{1}{2} \left(g^{\mu\rho} g^{\nu\sigma} + g^{\nu\rho} g^{\mu\sigma} - g^{\rho\sigma} g^{\mu\nu} \right) \square_x G_{\mu\nu\alpha\beta}(x-y) = -\frac{i}{2} \delta(x-y) \left(\delta_\alpha^\rho \delta_\beta^\sigma + \delta_\beta^\rho \delta_\alpha^\sigma \right) \quad \star$$

Let us just make few comments on this equation. [p. 34*****]

Why the Green's function has four indices?

In the presence of a source, the E.o.M. is schematically

$$\underbrace{\frac{1}{2} \left(g^{\mu\rho} g^{\nu\sigma} + g^{\mu\sigma} g^{\nu\rho} - g^{\mu\nu} g^{\rho\sigma} \right)}_{\equiv \mathcal{O}^{\rho\sigma\mu\nu}} \square h_{\mu\nu}(x) = T^{\rho\sigma}(x)$$

and the Green's function must map a rank-2 source at “y” into a rank-2 field at “x”

$$h_{\mu\nu}(x) = \int d^4y G_{\mu\nu\alpha\beta}(x-y) T^{\alpha\beta}(y) \quad \text{That's because it has four indices}$$

- { The indices $\alpha\beta$ tell which component of the source you feed in
- { The indices $\mu\nu$ tell which component of $h_{\mu\nu}$ you get out

Why $G_{\mu\nu\alpha\beta}(x, y)$ depends only on “ $x - y$ ”?

This is a consequence of invariance under space-time translations. Propagating from “y” to “x” depends only on the displacement between the two events, not on the absolute positions “x” and “y” separately.

What is the right-hand side?

The Green's function is defined by considering a unit point-like source. [p. 35*****]

So that we have

$$\mathcal{O}^{\rho\sigma\mu\nu} G_{\mu\nu\alpha\beta}(x-y) = -i \delta(x-y) \underbrace{\frac{1}{2} \left(\delta_\alpha^\rho \delta_\beta^\sigma + \delta_\beta^\rho \delta_\alpha^\sigma \right)}$$

This is the **identity in the space of symmetric rank-2 tensors**

Added explanation (not in the original notes)

Why the right-hand side is not simply $\delta_\alpha^\rho \delta_\beta^\sigma$. A propagator must be the inverse of the kinetic operator *within the space it acts on* — here, the space of *symmetric* rank-2 tensors. The identity on that space is the symmetrized Kronecker pair

$$\mathbb{K}^{\rho\sigma}{}_{\alpha\beta} = \frac{1}{2} \left(\delta_\alpha^\rho \delta_\beta^\sigma + \delta_\beta^\rho \delta_\alpha^\sigma \right),$$

which satisfies $\mathbb{K}^{\rho\sigma}{}_{\alpha\beta} S^{\alpha\beta} = S^{\rho\sigma}$ for any symmetric S . Using the unsymmetrized $\delta_\alpha^\rho \delta_\beta^\sigma$ instead would be inconsistent, since it does not map symmetric tensors to themselves in a well-defined way.

The factor $-i$. Convention: it is what makes G come out as the *Feynman propagator*, matching the $-ig_{\mu\nu}/(p^2 + i\epsilon)$ of the photon used on p. 39*****.

This is exactly the QED calculation with more indices. There one solves $\square_x D_{\mu\nu}(x-y) = -ig_{\mu\nu}\delta(x-y)$ and gets $\tilde{D}_{\mu\nu} = -ig_{\mu\nu}/(p^2 + i\epsilon)$. Everything below is the same three steps — Fourier transform, ansatz, match coefficients — with the vector index promoted to a symmetric pair.

■ We solve the equation in Fourier space. We introduce

$$G_{\mu\nu\alpha\beta}(x-y) = \int \frac{d^4p}{(2\pi)^4} e^{-ip \cdot (x-y)} \tilde{G}_{\mu\nu\alpha\beta}(p)$$

$$\delta(x-y) = \int \frac{d^4p}{(2\pi)^4} e^{-ip \cdot (x-y)}$$

and eq. ★ becomes

$$\left(g^{\mu\rho}g^{\nu\sigma} + g^{\nu\rho}g^{\mu\sigma} - g^{\rho\sigma}g^{\mu\nu}\right)p^2 \tilde{G}_{\mu\nu\alpha\beta}(p) = i\left(\delta_\alpha^\rho\delta_\beta^\sigma + \delta_\alpha^\sigma\delta_\beta^\rho\right)$$

■ We solve the above equation.

We consider the following ansatz for the solution:

$$\tilde{G}_{\mu\nu\alpha\beta}(p) = A(p^2)\left(g_{\mu\alpha}g_{\nu\beta} + g_{\mu\beta}g_{\nu\alpha}\right) + B(p^2)g_{\mu\nu}g_{\alpha\beta}$$

This ansatz is motivated by Lorentz covariance and symmetry w.r.t. both pairs $\mu\nu$ and $\alpha\beta$.
[p. 36*****]

We substitute the ansatz in the equation, and work out the left-hand side in detail:

$$\begin{aligned} & \left(g^{\mu\rho}g^{\nu\sigma} + g^{\mu\sigma}g^{\nu\rho} - g^{\mu\nu}g^{\rho\sigma}\right)p^2 \left[A(p^2)(g_{\mu\alpha}g_{\nu\beta} + g_{\mu\beta}g_{\nu\alpha}) + B(p^2)(g_{\mu\nu}g_{\alpha\beta})\right] = \\ & = p^2 A(p^2) \left[\delta_\alpha^\rho\delta_\beta^\sigma + \delta_\beta^\rho\delta_\alpha^\sigma + \delta_\alpha^\sigma\delta_\beta^\rho + \delta_\beta^\sigma\delta_\alpha^\rho - g_{\alpha\beta}g^{\rho\sigma} - g_{\beta\alpha}g^{\rho\sigma}\right] \\ & \quad + p^2 B(p^2) \left(g^{\rho\sigma}g_{\alpha\beta} + g^{\sigma\rho}g_{\alpha\beta} - \underbrace{(g^{\mu\nu}g_{\mu\nu})}_{D} g^{\rho\sigma}g_{\alpha\beta}\right) = \end{aligned}$$

For fun, let's work in D space-time dimensions, such that $g^{\mu\nu}g_{\mu\nu} = D$

$$\begin{aligned} & = p^2 A(p^2) 2\left(\delta_\alpha^\rho\delta_\beta^\sigma + \delta_\alpha^\sigma\delta_\beta^\rho - g_{\alpha\beta}g^{\rho\sigma}\right) + p^2 B(p^2)(2-D)g^{\rho\sigma}g_{\alpha\beta} = \\ & = 2p^2 A(p^2) \left(\delta_\alpha^\rho\delta_\beta^\sigma + \delta_\alpha^\sigma\delta_\beta^\rho\right) - p^2 g_{\alpha\beta}g^{\rho\sigma} \left[2A(p^2) + (D-2)B(p^2)\right] \end{aligned}$$

These terms must equal the right-hand side

$$\stackrel{!}{=} i\left(\delta_\alpha^\rho\delta_\beta^\sigma + \delta_\alpha^\sigma\delta_\beta^\rho\right)$$

and, from the comparison, we get, solving for “A” and “B” [p. 37*****]

$$2A(p^2)p^2 \stackrel{!}{=} i \quad \longrightarrow \quad A(p^2) = \frac{i}{2p^2}$$

$$2A(p^2) = (2-D)B(p^2) \quad \longrightarrow \quad B(p^2) = \frac{i}{p^2(2-D)}$$

■ We regulate the singularity as $p^2 \rightarrow 0$ with the Feynman $i\varepsilon$ prescription.

We arrive at the expression

$$\begin{aligned} \tilde{G}_F^{\mu\nu\rho\sigma}(p) &= \text{---}\overline{\text{---}}\text{---} \xrightarrow{p} = \\ &= \frac{i}{p^2 + i\varepsilon} \left[\frac{1}{2} (g^{\mu\rho} g^{\nu\sigma} + g^{\mu\sigma} g^{\nu\rho}) - \frac{1}{D-2} g^{\mu\nu} g^{\rho\sigma} \right] \end{aligned}$$

graviton propagator in D space-time dimensions

Added explanation (not in the original notes)

Reading off A and B . Matching the $\delta\delta$ structure gives $2Ap^2 = i$; matching the $g_{\alpha\beta}g^{\rho\sigma}$ structure, which must vanish since the right-hand side has none, gives $2A + (D-2)B = 0$, i.e. $2A = (2-D)B$. Substituting the ansatz back,

$$\tilde{G} = \frac{i}{2p^2} (g_{\mu\alpha}g_{\nu\beta} + g_{\mu\beta}g_{\nu\alpha}) + \frac{i}{p^2(2-D)} g_{\mu\nu}g_{\alpha\beta},$$

which is the boxed result once $1/(2-D) = -1/(D-2)$ is used.

In $D = 4$. The numerator becomes $\frac{1}{2} (g^{\mu\rho}g^{\nu\sigma} + g^{\mu\sigma}g^{\nu\rho} - g^{\mu\nu}g^{\rho\sigma})$ — the familiar textbook form. *The relative minus sign of the last term is what makes gravity attractive*, as the computation of Section 14 shows.

Why keep D general? Three reasons: it costs nothing; it makes the algebra's structure visible (the $1/(D-2)$ traces back to the trace part of the operator); and it is the form needed for dimensional regularization at loop level. Note the pole at $D = 2$: in two dimensions the graviton kinetic operator is not invertible at all, which is the field-theory shadow of the fact that the Einstein–Hilbert action is topological in 2D.

Sanity check on the counting. The propagator has no $p^\mu p^\nu / p^2$ terms because we fixed the harmonic gauge; a different gauge choice would add such terms, exactly as ξ -dependent $k^\mu k^\nu / k^2$ pieces appear in the photon propagator. Physical results do not depend on them.

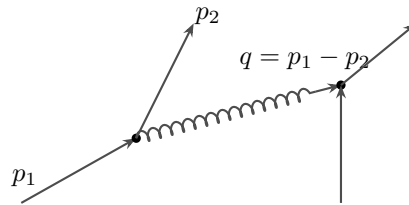
Part III — Forces from particle exchange

(pp. 37*****–47*****)

8 Forces from particle exchange: the setup [p. 37*****–38*****]

In perturbative QFT, forces are created by the exchange of virtual particles

The virtual particle that is exchanged carries some momentum q



↔ There is a change of momentum for both particles, therefore a force ($\vec{F} = d\vec{p}/dt$).

[p. 38*****]

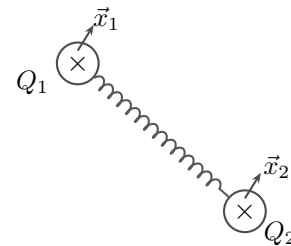
How to make the presence of this force manifest?

► The case of electromagnetic interactions

We consider the following situation. We have two sources located at position $\vec{x}_1$ and $\vec{x}_2$ described by static currents

$$J_{\text{cl},1}^\mu(\vec{x}) = Q_1 g^{\mu 0} \delta(\vec{x} - \vec{x}_1)$$

$$J_{\text{cl},2}^\mu(\vec{x}) = Q_2 g^{\mu 0} \delta(\vec{x} - \vec{x}_2)$$



They interact electromagnetically, with the theory

$$\mathcal{L} = -\frac{1}{4} F_{\mu\nu} F^{\mu\nu} - A_\mu (J_{\text{cl},1}^\mu + J_{\text{cl},2}^\mu)$$

↔ We compute the probability amplitude for creation, propagation and absorption of a virtual photon. In other words, we compute

Added explanation (not in the original notes)

The idea of the method in one sentence. Put two static charges into the vacuum and ask how much the vacuum energy shifts. That shift, as a function of the separation $r = |\vec{x}_1 - \vec{x}_2|$, is the interaction potential $V(r)$ — because the energy you must supply to bring the charges from infinity to distance r is precisely $E_0(r) - E_0(\infty)$.

Why sources rather than particles. Treating the charges as *external classical currents* rather than dynamical fields is a huge simplification: they do not recoil, they do not propagate, and the entire calculation reduces to one photon propagator connecting the two fixed points. Nothing physical is lost for the static potential.

$g^{\mu 0}$ and $\delta(\vec{x} - \vec{x}_i)$. $J^\mu \propto g^{\mu 0}$ means the current has only a charge density and no spatial current — the source sits still. The delta function makes it point-like. The whole configuration is time-independent, which is what makes the final answer a static potential.

9 The vacuum persistence amplitude [p. 39*****-40*****]

$$\langle 0 | U_I(+\infty, -\infty) | 0 \rangle = \langle 0 | \sum_{m=0}^{\infty} \frac{(-i)^m}{m!} \int d^4x_1 \cdots d^4x_m T[\mathcal{H}_{\text{int}}(x_1) \cdots \mathcal{H}_{\text{int}}(x_m)] | 0 \rangle$$

$\mathcal{H}_{\text{int}}(x) = A_\mu(x) [J_{\text{cl},1}^\mu(x) + J_{\text{cl},2}^\mu(x)]$ is the interaction Hamiltonian

$$\begin{aligned} \langle 0 | U_I(+\infty, -\infty) | 0 \rangle &= 1 + \cancel{(-i) \int d^4x \langle 0 | T(\mathcal{H}_{\text{int}}(x)) | 0 \rangle} \\ &\quad \text{vanishes since } \langle 0 | A_\mu(x) | 0 \rangle = 0 \\ &\quad + \frac{(-i)^2}{2} \int d^4x d^4y \langle 0 | T[\mathcal{H}_{\text{int}}(x) \mathcal{H}_{\text{int}}(y)] | 0 \rangle + \cdots \\ &= 1 + (-i)^2 \int d^4x d^4y \langle 0 | T[A_\mu(x) J_{\text{cl},1}^\mu(x) A_\nu(y) J_{\text{cl},2}^\nu(y)] | 0 \rangle + \cdots \\ &= 1 - \int d^4x d^4y \underbrace{J_{\text{cl},1}^\mu(x) J_{\text{cl},2}^\nu(y)}_{\text{classical sources}} \underbrace{\langle 0 | T[A_\mu(x) A_\nu(y)] | 0 \rangle}_{\text{photon Feynman propagator}} + \cdots \\ &= 1 - \int d^4x d^4y J_{\text{cl},1}^\mu(x) J_{\text{cl},2}^\nu(y) \int \frac{d^4p}{(2\pi)^4} e^{-ip \cdot (x-y)} \frac{(-ig_{\mu\nu})}{p^2 + i\varepsilon} = \end{aligned}$$

The classical sources go outside the expectation value; what is left is the photon Feynman propagator.

[p. 40*****]

$$\begin{aligned} &= 1 + iQ_1 Q_2 \int d^4x d^4y \delta(\vec{x} - \vec{x}_1) \delta(\vec{y} - \vec{x}_2) \int \frac{d^4p}{(2\pi)^4} g_{00} \frac{e^{-ip \cdot (x-y)}}{p^2 + i\varepsilon} \\ &= 1 + iQ_1 Q_2 \underbrace{\int d^4x d^4y \delta(\vec{x} - \vec{x}_1) \delta(\vec{y} - \vec{x}_2)}_{\text{we integrate}} \int \frac{d^4p}{(2\pi)^4} \frac{e^{-ip^0(x^0-y^0) + i\vec{p} \cdot (\vec{x}-\vec{y})}}{(p^0)^2 - |\vec{p}|^2 + i\varepsilon} \\ &= 1 + iQ_1 Q_2 \int dx^0 dy^0 \int \frac{d^3\vec{p} dp^0}{(2\pi)^4} \frac{e^{-ip^0(x^0-y^0)} e^{i\vec{p} \cdot (\vec{x}_1 - \vec{x}_2)}}{(p^0)^2 - |\vec{p}|^2 + i\varepsilon} \\ &\quad \text{we use } \int dx^0 e^{-ip^0 x^0} = (2\pi) \delta(p^0) \\ &= 1 + iQ_1 Q_2 \int dy^0 \int \frac{d^3\vec{p} dp^0}{(2\pi)^3} \frac{e^{ip^0 y^0} e^{i\vec{p} \cdot (\vec{x}_1 - \vec{x}_2)} \delta(p^0)}{(p^0)^2 - |\vec{p}|^2 + i\varepsilon} \\ &= 1 + iQ_1 Q_2 \underbrace{\left(\int dy^0 \right)}_{\text{diverges}} \int \frac{d^3\vec{p}}{(2\pi)^3} \frac{e^{i\vec{p} \cdot (\vec{x}_1 - \vec{x}_2)}}{-|\vec{p}|^2 + i\varepsilon} \end{aligned}$$

This integral diverges, but this is simply because we are integrating between $-\infty$ and $+\infty$ in time domain

We regularize the integral over y^0 by considering a finite time interval between $[-\frac{T}{2}, +\frac{T}{2}]$.

After all, what we are computing is the quantity [p. 41*****]

$$\langle 0|U_I(+\infty, -\infty)|0\rangle = \lim_{T \rightarrow \infty} \langle 0|U_I(\frac{T}{2}, -\frac{T}{2})|0\rangle$$

We thus have

$$\langle 0|U_I(\frac{T}{2}, -\frac{T}{2})|0\rangle = 1 + iQ_1Q_2T \int \frac{d^3\vec{p}}{(2\pi)^3} \frac{e^{i\vec{p}\cdot(\vec{x}_1-\vec{x}_2)}}{-|\vec{p}|^2 + i\varepsilon}$$

where we are interested in the limit in which T is very large

Added explanation (not in the original notes)

Bookkeeping of the factor 2. The second-order term carries $1/2!$, but the product $\mathcal{H}_{\text{int}}(x)\mathcal{H}_{\text{int}}(y)$ contains *two* cross terms ($J_1 \cdot J_2$ and $J_2 \cdot J_1$) which are equal after relabelling. The 2 cancels the $1/2!$, which is why the line with $(-i)^2$ appears without the $\frac{1}{2}$. The self-interaction terms J_1J_1 and J_2J_2 are dropped: they are r -independent self-energies and do not contribute to the force.

Where the T comes from. With static sources nothing depends on y^0 , so $\int dy^0 = T$ — an overall factor of the (regulated) total time. The amplitude therefore grows linearly in T , which is precisely the signature of a constant energy shift: a state of energy E_0 evolving for time T picks up $e^{-iE_0T} \approx 1 - iE_0T$. Comparing with the boxed result already gives E_0 — and the next pages make that comparison rigorous.

Why g_{00} appears. Both currents point along $\mu = 0$, so contracting with the photon propagator's $g_{\mu\nu}$ leaves $g_{00} = +1$. The sign of this single number is what will decide whether like charges attract or repel — and it is the analogous contraction for gravity (Section 14) that flips the answer.

10 The same quantity computed from the spectrum [p. 41*****-44*****]

Let's try to compute this quantity in a different way. We use the fact that the evolution operator in the interaction picture can be written as follows

$$U_I(t_2, t_1) = e^{iH_0t_2} e^{-iH(t_2-t_1)} e^{-iH_0t_1}$$

H : Hamiltonian of the fully interacting theory

H_0 : Hamiltonian of the free theory

$|0\rangle$ is the vacuum of the free theory, eigenstate of H_0 with vanishing eigenvalue, $H_0|0\rangle = 0$.

We thus write the expectation value

$$\langle 0|U_I(\frac{T}{2}, -\frac{T}{2})|0\rangle = \langle 0|e^{iH_0T/2} e^{-iH(T/2+T/2)} e^{iH_0T/2}|0\rangle = \langle 0|e^{-iHT}|0\rangle$$

[p. 42*****]

$$\langle 0|U_I(\frac{T}{2}, -\frac{T}{2})|0\rangle = \langle 0|e^{-iHT}|0\rangle$$



$\hookrightarrow$ We now insert a resolution of the identity, that we formally write as $\mathbb{1} = \sum_{\alpha} |\alpha\rangle\langle\alpha|$

Comment: Formally, $|\alpha\rangle$ are a complete set of eigenstates of the fully interacting Hamiltonian H :

$$H|\alpha\rangle = \alpha|\alpha\rangle$$

When possible, they are chosen so to diagonalize any other set of commuting conserved charges (momentum, spin, ...) in the interacting theory.

However, in an interacting theories we usually do not know their explicit form. In many arguments, such explicit form is not needed. What matters is that they **form a complete set** and that they are **eigenstates of the full Hamiltonian**.

Back to ★, we thus have [p. 43*****]

$$\langle 0|U_I(\frac{T}{2}, -\frac{T}{2})|0\rangle = \sum_{\alpha} \langle 0|e^{-iHT}|\alpha\rangle\langle\alpha|0\rangle = \sum_{\alpha} e^{-iE_{\alpha}T} |\langle\alpha|0\rangle|^2$$

$$\langle 0|U_I(\frac{T}{2}, -\frac{T}{2})|0\rangle = \sum_{\alpha} e^{-iE_{\alpha}T} |\langle\alpha|0\rangle|^2$$

As before, we are interested in the limit $T \rightarrow \infty$. However, as it is, the limit is problematic. The T -dependence only enters via a complex phase and, if we take the limit $T \rightarrow \infty$, the limit does not select any of the states in the infinite sum.

↪ We need to regulate the sum. We define the above sum by *analytic continuation*. We **give the time a small negative imaginary part**

$$T \longrightarrow T - i\epsilon, \quad \epsilon > 0$$

This produces in the exponential a damping factor

$$e^{-iE_{\alpha}T} \longrightarrow e^{-iE_{\alpha}(T-i\epsilon)} = e^{-iE_{\alpha}T} \underbrace{e^{-\epsilon E_{\alpha}}}_{\text{damping factor}}$$

This damping factor makes the high energy states exponentially suppressed. [p. 44*****]

Consequently

$$\langle 0|U_I(\frac{T}{2}, -\frac{T}{2})|0\rangle = \sum_{\alpha} e^{-iE_{\alpha}T} e^{-\epsilon E_{\alpha}} |\langle\alpha|0\rangle|^2$$

The sum is dominated by the state with the lowest energy in the limit in which T is very large.

We thus find

$$\langle 0|U_I(\frac{T}{2}, -\frac{T}{2})|0\rangle = e^{-iE_0T} e^{-\epsilon E_0} |\langle\Omega|0\rangle|^2$$

We arrive at

$$\begin{aligned} \langle 0|U_I(\frac{T}{2}, -\frac{T}{2})|0\rangle &= 1 + iQ_1Q_2T \int \frac{d^3\vec{p}}{(2\pi)^3} \frac{e^{i\vec{p}\cdot(\vec{x}_1-\vec{x}_2)}}{-|\vec{p}|^2 + i\epsilon} \\ &= e^{-iE_0T} e^{-\epsilon E_0} |\langle\Omega|0\rangle|^2 \end{aligned}$$

We use the following short cut. We take the log on both sides

$$\underbrace{\log \left(1 + iQ_1 Q_2 T \int \frac{d^3 \vec{p}}{(2\pi)^3} \frac{e^{i\vec{p} \cdot (\vec{x}_1 - \vec{x}_2)}}{|\vec{p}|^2 + i\epsilon} \right)}_{\text{we expand the log}} = -iE_0 T + \underbrace{\log \left(e^{-\epsilon E_0} |\langle \Omega | 0 \rangle|^2 \right)}_{\text{we neglect this term that does not depend on } T}$$

Added explanation (not in the original notes)

The strategy, stated plainly. The *same* object $\langle 0 | U_I(T/2, -T/2) | 0 \rangle$ has now been computed twice: once by Feynman diagrams (one photon exchange), once by spectral decomposition. Equating the two extracts E_0 — the energy of the true ground state $|\Omega\rangle$ *in the presence of the sources*. That is the trick.

Why $T \rightarrow T - i\epsilon$ is legitimate. It is a Wick rotation in disguise: giving the time an infinitesimal negative imaginary part turns the oscillating phases into decaying exponentials, so the sum converges and the ground state dominates. It is the same $i\epsilon$ that defines the Feynman propagator, and the same device that lets Euclidean path integrals project onto the vacuum as $T \rightarrow \infty$. Nothing physical depends on ϵ ; it is removed at the end.

Why the neglected term really is negligible. $\log(e^{-\epsilon E_0} |\langle \Omega | 0 \rangle|^2)$ is a constant — no T in it. Since we match the coefficients of the term *linear in T* on both sides, it drops out. Expanding the left-hand side, $\log(1+x) \approx x$ for small x , and comparing the $O(T)$ terms gives E_0 directly.

11 The vacuum energy, and the sign of the potential [p. 45*****]

We get

$$E_0 = Q_1 Q_2 \int \frac{d^3 \vec{p}}{(2\pi)^3} \frac{e^{i\vec{p} \cdot (\vec{x}_1 - \vec{x}_2)}}{|\vec{p}|^2 - i\epsilon}$$

energy of the vacuum in the presence of the two interacting sources

Since, in the absence of the sources, we started from a vacuum with no sources and zero energy, we interpret E_0 as **the energy that we gain or lose by bringing the two charges together from infinity** (potential energy).

↔ If the presence of the two sources **lowers** the vacuum energy compared to when they are far apart, it's energetically favorable to bring them closer: *attractive potential*.

$$E_0 < 0 \longrightarrow \text{attractive potential}$$

↔ If the presence of the two sources **raises** the vacuum energy, it's energetically unfavorable to bring them closer: *repulsive potential*.

$$E_0 > 0 \longrightarrow \text{repulsive potential}$$

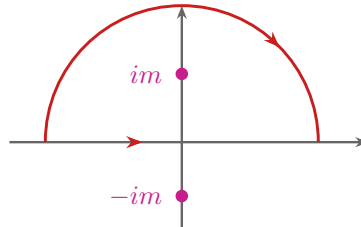
12 Computing the integral: the Coulomb potential [p. 46*****-47*****]

We compute explicitly the integral. We introduce

$$\vec{x}_1 - \vec{x}_2 \equiv \vec{r}, \quad r \equiv |\vec{r}|; \quad p \equiv |\vec{p}|$$

$$\begin{aligned}
 E_0 &= Q_1 Q_2 \int \frac{dp \, p^2 (2\pi) d(\cos \theta)}{(2\pi)^3} \frac{e^{irp \cos \theta}}{p^2 - i\varepsilon} = \\
 &= \frac{Q_1 Q_2}{ir} \int_{-\infty}^{+\infty} \frac{dp \, p}{4\pi^2} \frac{e^{irp}}{p^2 - i\varepsilon} \stackrel{(*)}{=} \frac{Q_1 Q_2}{ir} \int_{-\infty}^{+\infty} \frac{dp \, p}{4\pi^2} \frac{e^{irp}}{p^2 + m^2} =
 \end{aligned}$$

(*) It is instructive to compute this integral by introducing as regulator a non-vanishing photon mass, and take the massless limit at the end



$$\begin{aligned}
 &= \frac{Q_1 Q_2}{ir} \frac{1}{4\pi^2} (2\pi i) \operatorname{Res}_{z=im} \frac{z e^{irz}}{(z + im)(z - im)} = \\
 &= \frac{Q_1 Q_2}{4\pi^2 ir} (2\pi i) \frac{im e^{-rm}}{2im} = \frac{Q_1 Q_2 e^{-rm}}{4\pi r}
 \end{aligned}$$

$$E_0 = \frac{Q_1 Q_2 e^{-rm}}{4\pi r} \xrightarrow{m \rightarrow 0} \frac{Q_1 Q_2}{4\pi r}$$

We learn two interesting things: [p. 47*****]

1) The exchange of a massless photon correctly reproduces the Coulomb potential

$$\hookrightarrow \begin{cases} \text{equal-sign charges repel } (E_0 > 0) \\ \text{opposite-sign charges attract } (E_0 < 0) \end{cases}$$

2) If the photon were massive, we would get

$$E_0 = \frac{Q_1 Q_2 e^{-mr}}{4\pi r}$$

short-range potential, with a characteristic range of the order of the Compton wavelength of the particle mediating the interaction.

Added explanation (not in the original notes)

The contour integral, step by step. After doing the angular integral, $\int_{-1}^1 d(\cos \theta) e^{irp \cos \theta} = \frac{2 \sin(rp)}{rp}$, one rewrites the result as a full-line integral of $p e^{irp} / (p^2 + m^2)$ divided by ir . Closing the contour in the *upper* half-plane is allowed because e^{irp} decays there for $r > 0$; the only enclosed pole is $p = +im$, with residue $\frac{im e^{-rm}}{2im} = \frac{e^{-rm}}{2}$. Assembling the factors gives $Q_1 Q_2 e^{-rm} / (4\pi r)$.

The mass as a regulator. Replacing $-i\varepsilon \rightarrow +m^2$ moves the poles off the real axis in a controlled way and makes the integral manifestly convergent. Setting $m \rightarrow 0$ at the end recovers Coulomb. But the intermediate result is physics in its own right: this is the **Yukawa poten-**

tial, with range $1/m$ (restoring units, the Compton wavelength $\hbar/mc$). It is how Yukawa predicted the pion's mass from the range of the nuclear force.

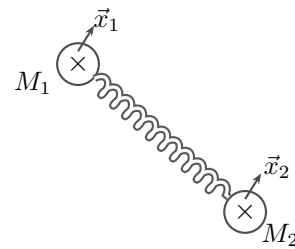
The lesson about signs. For the photon, the two like charges give $Q_1 Q_2 > 0$, hence $E_0 > 0$, hence repulsion. Nothing in the derivation prejudged this — it came from $g_{00} = +1$ appearing once in the contraction of the two currents with the propagator. *Vector exchange makes like charges repel.* Scalar exchange, and (as we are about to see) tensor exchange, do the opposite. This is the deep reason that gravity, alone among the fundamental forces, is universally attractive.

13 The spin-2 mediated interaction [p. 47*****]

Consider the two sources described by the energy-momentum tensor

$$T_{\text{cl},1}^{\mu\nu}(\vec{x}) = M_1 g^{\mu 0} g^{\nu 0} \delta(\vec{x} - \vec{x}_1)$$

$$T_{\text{cl},2}^{\mu\nu}(\vec{x}) = M_2 g^{\mu 0} g^{\nu 0} \delta(\vec{x} - \vec{x}_2)$$



They interact with

$$\mathcal{L} = \mathcal{L}_0 - \frac{\kappa}{2} h_{\mu\nu} (T_{\text{cl},1}^{\mu\nu} + T_{\text{cl},2}^{\mu\nu})$$

We use the same coupling κ since it is **universal**

[The scan ends here, with the spin-2 exchange set up but not yet computed.]

Added explanation (not in the original notes)

Carrying the calculation through — this is the payoff of the whole lecture, so it is worth the five lines the scan does not contain. Repeat the steps of Sections 10–12 with the photon propagator replaced by the graviton propagator of p. 37***** and the currents by the energy-momentum tensors:

$$E_0 = -\frac{\kappa^2}{4} T_1^{\mu\nu} \left[\frac{1}{2} (g_{\mu\rho} g_{\nu\sigma} + g_{\mu\sigma} g_{\nu\rho}) - \frac{1}{D-2} g_{\mu\nu} g_{\rho\sigma} \right] T_2^{\rho\sigma} \int \frac{d^3\vec{p}}{(2\pi)^3} \frac{e^{i\vec{p}\cdot\vec{r}}}{|\vec{p}|^2}.$$

With only the 00 components non-zero, the tensor contraction in $D = 4$ gives

$$\underbrace{\frac{1}{2} (g_{00} g_{00} + g_{00} g_{00})}_{=1} - \underbrace{\frac{1}{2} g_{00} g_{00}}_{=1/2} = \frac{1}{2},$$

a *positive* number, while the overall sign in front differs from the electromagnetic case. The momentum integral is the same one computed on p. 46*****, giving $1/(4\pi r)$. The result is

$$E_0 = -\frac{\kappa^2}{32\pi} \frac{M_1 M_2}{r} \equiv -G \frac{M_1 M_2}{r} \quad \text{with} \quad \kappa^2 = 32\pi G.$$

Three things to notice.

1. $E_0 < 0$ **always**: masses are positive, so the potential is *always attractive*, with no analogue of opposite charges. This traces directly to the numerator structure of the tensor

propagator — specifically to the $-\frac{1}{D-2}g^{\mu\nu}g^{\rho\sigma}$ term, which subtracts less than the first term adds.

2. **The general rule.** Even-spin exchange (scalar, tensor) gives attraction between like sources; odd-spin exchange (vector) gives repulsion between like charges. Each index of the polarization contributes a factor (-1) in the contraction of like sources.
3. **Newton emerges**, with G fixed in terms of κ . Combined with $[\kappa] = [M]^{-1}$ from Lecture 17, this identifies the Planck scale. So the programme that began in Lecture 16 with “take a symmetric rank-2 tensor and forbid ghosts” has now produced, with no geometric input whatsoever: the linearized Einstein equations, the equivalence principle, and Newton’s law of gravitation.

One notational point. The coupling here is written $-\frac{\kappa}{2}h_{\mu\nu}T^{\mu\nu}$ whereas Lecture 17 (p. 15*****) wrote $-\kappa h_{\mu\nu}T^{\mu\nu}$. This is only a redefinition of κ by a factor 2 and affects nothing but the numerical relation between κ and G ; different textbooks make different choices.

Summary of Lecture 18 — the logic in seven lines

1. Gauge invariance makes the graviton kinetic operator non-invertible, so a gauge must be fixed: the **harmonic gauge** $\partial_\mu \bar{h}^{\mu\nu} = 0$, with $\bar{h}_{\mu\nu} = h_{\mu\nu} - \frac{1}{2}g_{\mu\nu}h$.
2. The gauge is always reachable ($\square\xi^\nu = -\partial_\mu \bar{h}^{\mu\nu}$ always has solutions) and leaves a **residual freedom** with $\square\xi'^\mu = 0$: $10 - 4 - 4 = 2$ DOF.
3. In this gauge the equation of motion collapses to $\square\bar{h}^{\rho\sigma} = 0$.
4. The kinetic operator is $\mathcal{O}^{\rho\sigma\mu\nu} = \frac{1}{2}(g^{\mu\rho}g^{\nu\sigma} + g^{\mu\sigma}g^{\nu\rho} - g^{\mu\nu}g^{\rho\sigma})\square$; its Green’s function must be the identity *on symmetric rank-2 tensors*, hence four indices and a symmetrized $\delta\delta$ on the right-hand side.
5. A Lorentz-covariant ansatz plus index matching gives the **graviton propagator** $\frac{i}{p^2+i\varepsilon}[\frac{1}{2}(g^{\mu\rho}g^{\nu\sigma} + g^{\mu\sigma}g^{\nu\rho}) - \frac{1}{D-2}g^{\mu\nu}g^{\rho\sigma}]$.
6. Forces come from exchange: put two static sources in the vacuum, compute $\langle 0|U_I|0\rangle$ twice — by one-boson exchange and by spectral decomposition — and read off the shift E_0 in the vacuum energy. $E_0 < 0$ is attraction, $E_0 > 0$ repulsion.
7. For the photon this gives Coulomb, $E_0 = Q_1Q_2/(4\pi r)$ (and Yukawa $\propto e^{-mr}/r$ if the mediator is massive). For the graviton it gives $E_0 = -GM_1M_2/r$: *always attractive*.